

Palash Shukla

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EDUCATION

Mayoor School, Noida

Central Board of Secondary Education (CBSE) - Grade 11

Noida, India

Expected May 2027

LEADERSHIP & EXPERIENCE

Lead Organizer & Technical Director

Battle of Titans (BOT) Tech Fest

May 2026 – Present

Noida, India

- Scouting and recruiting a cross-functional team of top-tier student talent to execute the technical architecture and event logistics for a premier regional tech fest.
- Architecting a scalable full-stack registration portal using Next.js and Firebase, designed to handle high-concurrency traffic bursts for upcoming events.
- Directing school club expansion initiatives, establishing recruitment pipelines to onboard junior developers into technical workflows.

Independent Developer

Hack Club YSWS Program

Jun. 2024 – Present

Remote

- Participated in the “You Ship, We Ship” initiative, developing and deploying modular Python logic engines and CLI applications.
- Established a Figma-to-Tailwind design-to-code pipeline, complete with plugin integrations, preparing collaborative infrastructure for future team sprints.

PROJECTS

OptionPricer | [GitHub](#) | [PyPI](#) | *Cython, Python, OpenMP, NumPy*

Jan. 2026 – Present

- Developing a high-performance quantitative finance library published on PyPI, utilizing Monte Carlo simulations with active integration of GARCH modeling for Black-Scholes Greeks calculation.
- Leveraging **Cython AOT compilation** to bypass the Global Interpreter Lock (GIL) to optimize nested parameter loops.
- Implementing **OpenMP parallelism** to distribute computational loads, targeting reduced execution time in high-frequency inner loops.

Systems Security & Exploit Development | *x86 Assembly, C, Linux*

Aug. 2025 – Present

- Attained a peak **Global Rank of #66** on the pwn.college weekly leaderboard, independently engineering exploits to solve advanced challenges in binary exploitation.
- Bypassed kernel mitigations, including **KASLR**, by writing custom exploits from scratch utilizing memory corruption and pointer leak techniques.
- Maintained a consistent podium track record (1st, 2nd, and 3rd place finishes) across multiple global and regional Capture The Flag (CTF) and Cryptic Hunt competitions.

HONORS & ACHIEVEMENTS

Global Rankings: Ranked **#66** on the pwn.college weekly leaderboard and **Sub-5000** on the monthly leaderboard for vulnerability research (May 2026).

Competitive Performance: Secured consistent podium placements (Top 3) in multiple high-stakes CTF and Cryptic Hunt events, including a 2nd Place finish in the “Surprise” hunt.

TECHNICAL SKILLS

Systems & Security: Kernel Security (KASLR), x86-64 Assembly, Cython, OpenMP, GDB, Wireshark, Linux Internals
Development & Design: Next.js, React, Firebase, Tailwind CSS, Python (NumPy), Git, Docker, Figma (UI/UX)